Mathematics II

Marcos Bujosa

Universidad Complutense de Madrid

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You can find the last version of these course materials at

https://mbujosab.github.io/MatematicasII/

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1 Highlights of Lesson 1

Highlights of Lesson 1

- Vector and matrix operations
  - Addition and scalar multiplication
  - Some properties of these operations

2 Vectors in $\mathbb{R}^n$

A vector in $\mathbb{R}^n$ is an ordered list of n real numbers.

Example

$\boldsymbol{v} \in \mathbb{R}^3$: First component: 5, the second: 1 and the third: 10

$$\boldsymbol{v} = \begin{cases} v_1 = 5 \\ v_2 = 1 \\ v_3 = 10 \end{cases} \quad ; \qquad \boldsymbol{v} = \begin{pmatrix} 5, & 1, & 10, \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ 10 \end{pmatrix}.$$

Notation

- $\boldsymbol{a}$, $\boldsymbol{x}$, $\boldsymbol{0}$
- $\operatorname{elem}_3(\boldsymbol{v}) \equiv {}_{3|}\boldsymbol{v} \equiv \boldsymbol{v}_{|3} \equiv v_3 = 10$

A parenthesis around a list of numbers denotes a vector.

3 Basic operations with vectors

Vector addition: $(\boldsymbol{a}+\boldsymbol{b})_{|i} = \boldsymbol{a}_{|i} + \boldsymbol{b}_{|i}$

$$\boldsymbol{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \text{ and } \boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \quad \text{add to} \quad \boldsymbol{a}+\boldsymbol{b} = \begin{pmatrix} a_1+b_1 \\ a_2+b_2 \end{pmatrix}.$$

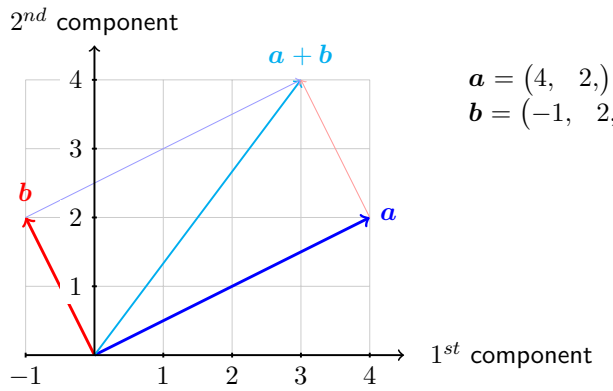
Scalar multiplication: $(\lambda\boldsymbol{a})_{|i} = \lambda(\boldsymbol{a}_{|i})$

$$2\boldsymbol{a} = \begin{pmatrix} 2a_1 \\ 2a_2 \end{pmatrix} \qquad \text{and} \qquad (-1)\boldsymbol{a} = \begin{pmatrix} -a_1 \\ -a_2 \end{pmatrix} \equiv -\boldsymbol{a}$$

(Hence, the operator " $|i$ " is linear)

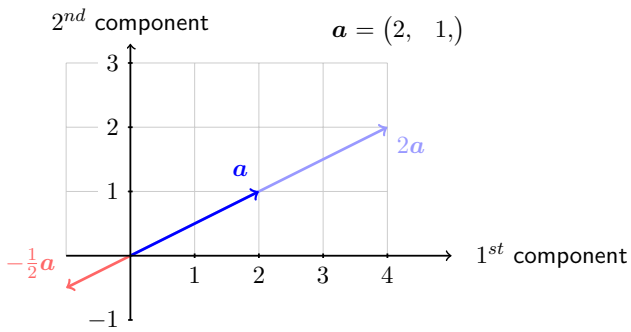
$\boldsymbol{a}$ and $\boldsymbol{b}$ (with n components) are equal when:
$\boldsymbol{a}_{|i} = \boldsymbol{b}_{|i}, \quad i = 1:n.$

4 Vector addition



$\boldsymbol{a} = (4, \quad 2,)$
$\boldsymbol{b} = (-1, \quad 2,)$

5 Scalar multiplication



What is the picture of all multiples of $\boldsymbol{a}$?
Is $\mathbf{0}$ a multiple of $\boldsymbol{a}$?

6 Addition and scalar multiplication

$$\boxed{(\boldsymbol{a}+\boldsymbol{b})_{|i} = \boldsymbol{a}_{|i} + \boldsymbol{b}_{|i}} \qquad \boxed{(\lambda\boldsymbol{a})_{|i} = \lambda(\boldsymbol{a}_{|i})}$$

Let us recall some properties of scalars

Scalars

1. $a+b = b+a$
2. $a+(b+c) = (a+b)+c$
3. $a+0 = a$
4. $a+(-a) = 0$
5. $ab = ba$
6. $a(b+c) = ab+ac$
7. $a(bc) = (ab)c$
8. $1a = a$

Vectors

1. $\boldsymbol{a}+\boldsymbol{b} = \boldsymbol{b}+\boldsymbol{a}$
2. $\boldsymbol{a}+(\boldsymbol{b}+\boldsymbol{c}) = (\boldsymbol{a}+\boldsymbol{b})+\boldsymbol{c}$
3. $\boldsymbol{a}+\mathbf{0} = \boldsymbol{a}$
4. $\boldsymbol{a}+(-\boldsymbol{a}) = \mathbf{0}$
5. $\lambda(\boldsymbol{a}+\boldsymbol{b}) = \lambda\boldsymbol{a}+\lambda\boldsymbol{b}$
6. $(\lambda+\eta)\boldsymbol{a} = \lambda\boldsymbol{a}+\eta\boldsymbol{a}$
7. $\lambda(\eta\boldsymbol{a}) = (\lambda\eta)\boldsymbol{a}$
8. $1\boldsymbol{a} = \boldsymbol{a}$

7 Matrices

A matrix in $\mathbb{R}^{m\times n}$ is an ordered list of n vectors in $\mathbb{R}^m$

Example

Three vectors in $\mathbb{R}^2$: $\boldsymbol{a} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$, $\boldsymbol{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ and $\boldsymbol{c} = \begin{pmatrix} 0 \\ 7 \end{pmatrix}$

$$\mathbf{A} = [\ \boldsymbol{a};\ \ \boldsymbol{b};\ \ \boldsymbol{c};\ ] = \begin{bmatrix} 4 & -1 & 0 \\ 2 & 2 & 7 \end{bmatrix} \neq [\ \boldsymbol{c};\ \ \boldsymbol{b};\ \ \boldsymbol{a};\ ]$$

Two vectors in $\mathbb{R}^3$: $\boldsymbol{x} = (4,\ \ -1,\ \ 0,)$ and $\boldsymbol{y} = (2,\ \ 2,\ \ 7,)$

$$\mathbf{B} = [\ \boldsymbol{x};\ \ \boldsymbol{y};\ ]$$

Notation

- $\mathbf{A}$, $\mathbf{B}$, $\mathbf{0}$
- $\underset{2\times 3}{\mathbf{A}}$, $\underset{3\times 2}{\mathbf{B}}$; $\qquad \underset{2\times 3}{\mathbf{A}} \neq \underset{3\times 2}{\mathbf{B}}$

A bracket around a vector list denotes a matrix.

8 More notation

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 7 & 0 & 3 \end{bmatrix}$$

Picking operators

- $\mathrm{elem}_{21}(\mathbf{A}) \ = \ {}_{2|}\mathbf{A}_{|1} \ = \ a_{21}: \qquad 7$
- $\mathrm{row}_1(\mathbf{A}) \ = \ {}_{1|}\mathbf{A} \ : \qquad (1,\ \ 2,\ \ 1,) = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$
- $\mathrm{col}_1(\mathbf{A}) \ = \ \mathbf{A}_{|1} \ : \qquad \begin{pmatrix} 1 \\ 7 \end{pmatrix} = (1,\ \ 7,)$

9 Basic operations with matrices

Matrix addition: $(\mathbf{A}+\mathbf{B})_{|j} = \mathbf{A}_{|j} + \mathbf{B}_{|j}$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \text{ add to } \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} \\ a_{21}+b_{21} & a_{22}+b_{22} \end{bmatrix}$$

Scalar multiplication: $(\lambda\mathbf{A})_{|j} = \lambda(\mathbf{A}_{|j})$

$$7\mathbf{A} = \begin{bmatrix} 7a_{11} & 7a_{12} \\ 7a_{21} & 7a_{22} \end{bmatrix} \text{ and } (-1)\mathbf{A} = \begin{bmatrix} -a_{11} & -a_{12} \\ -a_{21} & -a_{22} \end{bmatrix} = -\mathbf{A}.$$

(Hence, the operator " $|i$ " is linear)

$\mathbf{A}$ and $\mathbf{B}$ (with the same order) are equal when: $\mathbf{A}_{|j} = \mathbf{B}_{|j}$

10 Addition and scalar multiplication

$$\boxed{(\mathbf{A}+\mathbf{B})_{|j} = \mathbf{A}_{|j} + \mathbf{B}_{|j}}; \qquad \boxed{(\lambda\mathbf{A})_{|j} = \lambda(\mathbf{A}_{|j})}$$

Vectors

1. $\boldsymbol{a}+\boldsymbol{b} = \boldsymbol{b}+\boldsymbol{a}$
2. $\boldsymbol{a}+(\boldsymbol{b}+\boldsymbol{c}) = (\boldsymbol{a}+\boldsymbol{b})+\boldsymbol{c}$
3. $\mathbf{0}+\boldsymbol{a} = \boldsymbol{a}$
4. $\boldsymbol{a}+(-\boldsymbol{a}) = \mathbf{0}$
5. $\lambda(\boldsymbol{a}+\boldsymbol{b}) = \lambda\boldsymbol{a}+\lambda\boldsymbol{b}$
6. $(\lambda+\eta)\boldsymbol{a} = \lambda\boldsymbol{a}+\eta\boldsymbol{a}$
7. $\lambda(\eta\boldsymbol{a}) = (\lambda\eta)\boldsymbol{a}$
8. $1\boldsymbol{a} = \boldsymbol{a}$

Matrices

1. $\mathbf{A}+\mathbf{B} = \mathbf{B}+\mathbf{A}$
2. $\mathbf{A}+(\mathbf{B}+\mathbf{C}) = (\mathbf{A}+\mathbf{B})+\mathbf{C}$
3. $\mathbf{0}+\mathbf{A} = \mathbf{A}$
4. $\mathbf{A}+(-\mathbf{A}) = \mathbf{0}$
5. $\lambda(\mathbf{A}+\mathbf{B}) = \lambda\mathbf{A}+\lambda\mathbf{B}$
6. $(\lambda+\eta)\mathbf{A} = \lambda\mathbf{A}+\eta\mathbf{A}$
7. $\lambda(\eta\mathbf{A}) = (\lambda\eta)\mathbf{A}$
8. $1\mathbf{A} = \mathbf{A}$

11 Rewriting Rules

Distributive rules

$$(\boldsymbol{a}+\boldsymbol{b})_{|i} = \boldsymbol{a}_{|i} + \boldsymbol{b}_{|i} \qquad {}_{i|}(\boldsymbol{a}+\boldsymbol{b}) = {}_{i|}\boldsymbol{a} + {}_{i|}\boldsymbol{b}$$

$$(\mathbf{A}+\mathbf{B})_{|j} = \mathbf{A}_{|j} + \mathbf{B}_{|j} \qquad {}_{i|}(\mathbf{A}+\mathbf{B}) = {}_{i|}\mathbf{A} + {}_{i|}\mathbf{B}$$

In addition, if we allow $\lambda\boldsymbol{a} = \boldsymbol{a}\lambda$ and $\lambda\mathbf{A} = \mathbf{A}\lambda$, then we get

Associative rules (moving parentheses)

$$(\lambda\boldsymbol{b})_{|i} = \lambda(\boldsymbol{b}_{|i}) \qquad {}_{i|}(\boldsymbol{b}\lambda) = ({}_{i|}\boldsymbol{b})\lambda$$

$$(\lambda\mathbf{A})_{|j} = \lambda(\mathbf{A}_{|j}) \qquad {}_{i|}(\mathbf{A}\lambda) = ({}_{i|}\mathbf{A})\lambda$$

Scalar and operator interchange

$$(\boldsymbol{b}\lambda)_{|i} = (\boldsymbol{b}_{|i})\lambda \qquad {}_{i|}(\lambda\boldsymbol{b}) = \lambda({}_{i|}\boldsymbol{b})$$

$$(\mathbf{A}\lambda)_{|j} = (\mathbf{A}_{|j})\lambda \qquad {}_{i|}(\lambda\mathbf{A}) = \lambda({}_{i|}\mathbf{A})$$

Questions of the Lecture 1

You should always complete the exercises in the theoretical sections prior to each lecture

- Prove the properties of vector operations.
- Prove the properties of matrix operations.

(L-1) Question 1. Provide non-zero examples of 3 by 3 matrices of:

(a) A diagonal matrix: ${}_{i|}\mathbf{A}_{|j} = 0$ if $i \neq j$.
(b) A symmetric matrix: $\mathbf{A}_{|j} = {}_{j|}\mathbf{A}$.
(c) An upper triangular matrix: ${}_{i|}\mathbf{A}_{|j} = 0$ if $i > j$.
(d) A skew-symmetric matrix: ${}_{i|}\mathbf{A}_{|j} = -{}_{j|}\mathbf{A}_{|i}$.

(Strang, 1988, exercise 7 from section 1.4.)

1 Highlights of Lesson 2

Highlights of Lesson 2

- Dot product
- linear combinations
- The column picture of the geometry of linear equations

2 Dot product

$$\boldsymbol{x}\cdot\boldsymbol{y} = x_1y_1 + x_2y_2 + x_3y_3 + \cdots + x_ny_n = \sum_{i=1}^{n} x_iy_i.$$

Symmetric

$$\boldsymbol{x}\cdot\boldsymbol{y} = \boldsymbol{y}\cdot\boldsymbol{x}$$

Linear in the first argument

$$(a\boldsymbol{x})\cdot\boldsymbol{y} = a(\boldsymbol{x}\cdot\boldsymbol{y})$$

$$(\boldsymbol{x}+\boldsymbol{y})\cdot\boldsymbol{z} = \boldsymbol{x}\cdot\boldsymbol{z} + \boldsymbol{y}\cdot\boldsymbol{z}$$

Positive

$$\boldsymbol{x}\cdot\boldsymbol{x} \geq 0$$

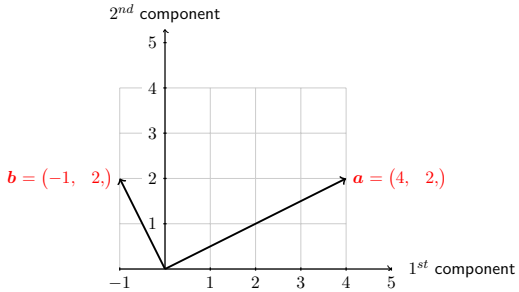
Definite

$$\boldsymbol{x}\cdot\boldsymbol{x} = 0 \Leftrightarrow \boldsymbol{x} = 0$$

3 Linear combinations

The sum of $x\boldsymbol{a}$ and $y\boldsymbol{b}$ is a linear combination of $\boldsymbol{a}$ and $\boldsymbol{b}$

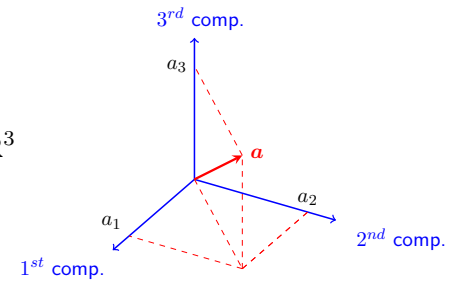
$$x\boldsymbol{a}+y\boldsymbol{b} = x\begin{pmatrix}4\\2\end{pmatrix}+y\begin{pmatrix}-1\\2\end{pmatrix} = [\boldsymbol{a};\ \boldsymbol{b};]\begin{pmatrix}x\\y\end{pmatrix} = \textbf{MATRIX}\times\boldsymbol{vector}$$



Is $\mathbf{0}$ a linear combination of $\boldsymbol{a}$ and $\boldsymbol{b}$? What is the picture of all linear combinations of $\boldsymbol{a}$ and $\boldsymbol{b}$?

4 Linear combinations in $\mathbb{R}^3$

$$\boldsymbol{a} = (a_1,\quad a_2,\quad a_3,) \in \mathbb{R}^3$$



What is the picture of all multiples of $\boldsymbol{a}$?
What is the picture of all linear combinations of two vectors in $\mathbb{R}^3$?
(linear combination)

5 Matrix times vector

$$\mathbf{A}\boldsymbol{b} = b_1\mathbf{A}_{|1} \quad + \quad b_2\mathbf{A}_{|2} \quad + \cdots + \quad b_n\mathbf{A}_{|n}$$

$$= b_1\begin{pmatrix}a_{11}\\ \vdots\\ a_{i1}\\ \vdots\\ a_{m1}\end{pmatrix} + b_2\begin{pmatrix}a_{12}\\ \vdots\\ a_{i2}\\ \vdots\\ a_{m2}\end{pmatrix} + \cdots + b_n\begin{pmatrix}a_{1n}\\ \vdots\\ a_{in}\\ \vdots\\ a_{mn}\end{pmatrix} = \begin{pmatrix}({}_{1|}\mathbf{A})\cdot\boldsymbol{b}\\ \vdots\\ ({}_{i|}\mathbf{A})\cdot\boldsymbol{b}\\ \vdots\\ ({}_{n|}\mathbf{A})\cdot\boldsymbol{b}\end{pmatrix}$$

Hence,

$$\boxed{{}_{i|}(\mathbf{A}\boldsymbol{b}) = ({}_{i|}\mathbf{A})\cdot\boldsymbol{b}}$$

if we omit the period, we can simply write: ${}_{i|}\mathbf{A}\boldsymbol{b}$

6 Matrix times vector

If $\boldsymbol{b} \in \mathbb{R}^n$ then $(\underset{m\times n}{\mathbf{A}}\ \boldsymbol{b}) \in \mathbb{R}^m$; where ${}_{i|}(\mathbf{A}\boldsymbol{b}) = ({}_{i|}\mathbf{A})\cdot\boldsymbol{b}$

Matrix times vector

1. $\mathbf{I}\boldsymbol{a} = \boldsymbol{a}$
2. $\mathbf{A}(\mathbf{I}_{|j}) = \mathbf{A}_{|j}$
3. $\mathbf{A}(\boldsymbol{b}+\boldsymbol{c}) = \mathbf{A}\boldsymbol{b} + \mathbf{A}\boldsymbol{c}$
4. $\mathbf{A}(\lambda\boldsymbol{b}) = \lambda(\mathbf{A}\boldsymbol{b})$
5. $\mathbf{A}(\lambda\boldsymbol{b}) = (\lambda\mathbf{A})\boldsymbol{b}$
6. $\mathbf{A}(\mathbf{B}\boldsymbol{c}) = \left[\mathbf{A}(\mathbf{B}_{|1});\ldots\quad \mathbf{A}(\mathbf{B}_{|n});\right]\boldsymbol{c}$
7. $(\mathbf{A}+\mathbf{B})\boldsymbol{c} = \mathbf{A}\boldsymbol{c} + \mathbf{B}\boldsymbol{c}$

Prove the above propositions (Book exercises: 14 to 18)
(follow the rewriting rules and properties of the dot product)

7 Example of linear system: 2 equations and 2 unknowns

$$\begin{cases} 2x - y = 0 \\ -x + 2y = 3 \end{cases}$$

$$\begin{bmatrix} & \\ & \end{bmatrix} \begin{pmatrix} \\ \end{pmatrix} = \begin{pmatrix} \\ \end{pmatrix}$$

$$\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$$

$$\underbrace{\mathbf{A}\boldsymbol{x}}_{\text{which linear combination}} \underbrace{= \boldsymbol{b}}_{\text{equals this vector?}}$$

$$x(\mathbf{A}_{|1}) + y(\mathbf{A}_{|2}) = \boldsymbol{b}$$

8 Geometry of linear systems: Linear combination of columns

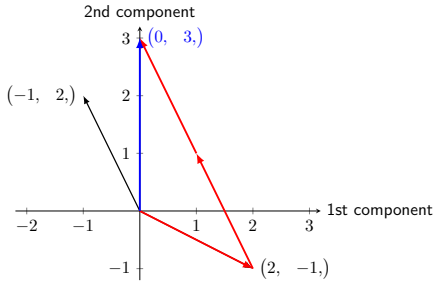
$$\begin{cases} 2x - y = 0 \\ -x + 2y = 3 \end{cases}$$

$$x\begin{pmatrix} \\ \end{pmatrix} + y\begin{pmatrix} \\ \end{pmatrix} = \begin{pmatrix} \\ \end{pmatrix}$$

9 2 equations and 2 unknowns: Column picture

$$x\begin{pmatrix} 2 \\ -1 \end{pmatrix} + y\begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$$

Which linear combination of $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ gives $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$?



What is the set of all possible combinations?

10 Example: 3 equations and 3 unknowns

$$\begin{cases} 2x - y \phantom{{}+4z} = 0 \\ -x + 2y - z = -1 \\ - 3y + 4z = 4 \end{cases}$$

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{pmatrix} \\ \\ \end{pmatrix} = \begin{pmatrix} \\ \\ \end{pmatrix}$$

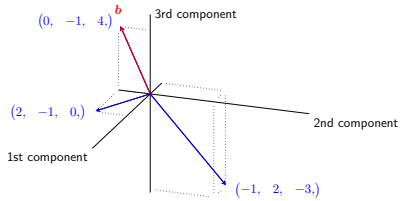
$$\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$$

$$\underbrace{\mathbf{A}\boldsymbol{x}}_{\text{which linear combination}} \underbrace{= \boldsymbol{b}}_{\text{equals this vector?}}$$

11 3 equations and 3 unknowns: Column picture

$$x\begin{pmatrix}2\\-1\\0\end{pmatrix}+y\begin{pmatrix}-1\\2\\-3\end{pmatrix}+z\begin{pmatrix}0\\-1\\4\end{pmatrix}=\begin{pmatrix}0\\-1\\4\end{pmatrix}$$

Which linear combination of the columns gives $\boldsymbol{b}$?



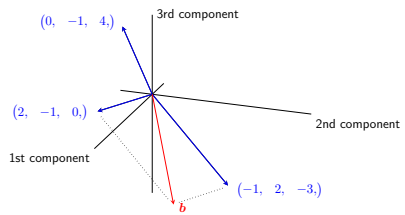
$$\left\{x= \quad ; \quad y= \quad ; \quad z= \quad \right\}$$

What happens if $\boldsymbol{b}$ is different?. . . let's see

12 3 equations and 3 unknowns: Column picture

$$x\begin{pmatrix}2\\-1\\0\end{pmatrix}+y\begin{pmatrix}-1\\2\\-3\end{pmatrix}+z\begin{pmatrix}0\\-1\\4\end{pmatrix}=\begin{pmatrix}1\\1\\-3\end{pmatrix}$$

Which linear combination of the columns gives this new $\boldsymbol{b}$?



$$\left\{x= \quad ; \quad y= \quad ; \quad z= \quad \right\}$$

13 What does Ax=b mean?

$\mathbf{A}\boldsymbol{x}$ is a *linear combination* of the columns of $\mathbf{A}$:

Example

$$\begin{bmatrix}2&5\\1&3\end{bmatrix}\begin{pmatrix}1\\2\end{pmatrix}=1\begin{pmatrix}2\\1\end{pmatrix}+2\begin{pmatrix}5\\3\end{pmatrix}=\begin{pmatrix}12\\7\end{pmatrix}$$

"$\mathbf{A}\boldsymbol{x}=\boldsymbol{b}$" is asking for a particular linear combination:

$$\begin{bmatrix}2&5\\1&3\end{bmatrix}\begin{pmatrix}\ \\ \ \end{pmatrix}=\begin{pmatrix}12\\7\end{pmatrix}$$

To solve linear systems we will first learn to transform coefficient matrices through elimination (next lessons)

Questions of the Lecture 2

You must complete the exercises from the corresponding sections of the book

(L-2) Question 1. Working a column at a time, compute the following products

(a)

$$\begin{bmatrix}4&1\\4&1\\6&1\end{bmatrix}\begin{pmatrix}1\\3\end{pmatrix}$$

(b)

$$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}\begin{pmatrix}0\\1\\0\end{pmatrix}$$

(c)

$$\begin{bmatrix}4&3\\6&6\\8&9\end{bmatrix}\begin{pmatrix}1/2\\1/3\end{pmatrix}$$

(Strang, 1988, exercise 2 from section 1.4.)

(L-2) Question 2. Can the three equations be solved simultaneously?

$$\begin{aligned} x + 2y &= 2 \\ x - y &= 2 \\ y &= 1. \end{aligned}$$

What happens if all right-hand sides are zero? Is there any non-zero choice of right-hand sides which allows the three equations to have a solution? How many non-zero choices have we?
(Strang, 1988, exercise 4 from section 1.2.)

(L-2) Question 3. Compute the product $\mathbf{A}\boldsymbol{x}$ with $\mathbf{A} = \begin{bmatrix} 3 & -6 & 0 \\ 0 & 2 & -2 \\ 1 & -1 & -1 \end{bmatrix}$ and $\boldsymbol{x} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$. For this matrix $\mathbf{A}$, find a solution vector $\boldsymbol{x}$ to the system $\mathbf{A}\boldsymbol{x} = \mathbf{0}$, with zeros on the right side of all three equations. Can you find more than one solution?
(Strang, 1988, exercise 5 from section 1.4.)

(L-2) Question 4. Suppose $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ has two solutions $\boldsymbol{v}$ and $\boldsymbol{w}$ (with $\boldsymbol{b} \neq \mathbf{0}$). Then show that $\frac{1}{2}(\boldsymbol{v} + \boldsymbol{w})$ is also a solution, although $\boldsymbol{v} + \boldsymbol{w}$ is not.

Hint
Use the following properties: $\mathbf{A}(\boldsymbol{b} + \boldsymbol{c}) = \mathbf{A}\boldsymbol{b} + \mathbf{A}\boldsymbol{c}$ and $\mathbf{A}(c\boldsymbol{b}) = c(\mathbf{A}\boldsymbol{b})$.

(L-2) Question 5. *"It is impossible for a system of linear equations to have exactly two solutions"*. Explain why (answering the next question):

(a) If $\boldsymbol{v}$ and $\boldsymbol{w}$ are two solutions, what is another one?

(Strang, 2003, exercise 19 from section 2.2.)

(L-2) Question 6. Draw $\boldsymbol{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\boldsymbol{w} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, along with $\boldsymbol{v} + \boldsymbol{w}$, $2\boldsymbol{v} + \boldsymbol{w}$, and $\boldsymbol{v} - \boldsymbol{w}$ in a plane (first component on the horizontal axis and second component on the vertical axis).

(L-2) Question 7. Draw the column picture of the following system with solution $x = 3$ and $y = -1$.

$$\begin{cases} 2x + y = 5 \\ x - 3y = 6 \end{cases}$$

(L-2) Question 8. Draw the column picture of the following system.

$$\begin{cases} 2x - y = 3 \\ x + y = 1 \end{cases} \quad ; \quad \left(\text{the solution is :} \quad x = 1 + \frac{1}{3}, \quad y = -\frac{1}{3} \right).$$

Don't stop doing the exercises from the book.

1 Highlights of Lesson 3

Highlights of Lesson 3

- Matrix multiplication: $(\mathbf{AB})_{|j} = \mathbf{A}(\mathbf{B}_{|j})$
  - Properties
- Transpose of a matrix
- $\mathbf{A}\boldsymbol{x}$ and $\boldsymbol{x}\mathbf{A}$ (linear combinations)
- Other ways to compute the product
- Transpose of $\mathbf{AB}$

2 Matrix multiplication (by **columns**)

Column j of $\big(\underset{m\times p}{\mathbf{A}}$ times $\underset{p\times n}{\mathbf{B}}\big)$ is:

$$\boxed{(\mathbf{AB})_{|j} = \mathbf{A}(\mathbf{B}_{|j})} \quad \longrightarrow \quad \mathbf{AB}_{|j}$$

Each column of **AB** is a linear combination of the p columns of **A**

Example

$$\begin{bmatrix} 2 & 1 \\ 3 & 8 \\ 4 & 9 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \mathbf{A}(\mathbf{B}_{|1}); & \mathbf{A}(\mathbf{B}_{|2}); \end{bmatrix}$$

$$= \begin{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 8 \\ 4 & 9 \end{bmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}; & \begin{bmatrix} 2 & 1 \\ 3 & 8 \\ 4 & 9 \end{bmatrix} \begin{pmatrix} 6 \\ 0 \end{pmatrix}; \end{bmatrix} = \begin{bmatrix} 3 & 12 \\ 11 & 18 \\ 13 & 24 \end{bmatrix}$$

3 Matrix multiplication properties

MATRIX $\times$ MATRIX $=$ MATRIX

1. $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$ remember $\mathbf{A}(\mathbf{B}\boldsymbol{c}) = \big[\mathbf{A}(\mathbf{B}_{|1}); \ldots \mathbf{A}(\mathbf{B}_{|n});\big]\,\boldsymbol{c}$
2. $(\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$.
3. $\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB} + \mathbf{AC}$.
4. $\mathbf{A}(\lambda\mathbf{B}) = (\lambda\mathbf{A})\mathbf{B} = \lambda(\mathbf{AB})$.
5. $\mathbf{IA} = \mathbf{A}$.
6. $\mathbf{AI} = \mathbf{A}$.

4 Transposing a matrix

Transpose

$$(\text{column } i \text{ of } \mathbf{A}^\mathsf{T}) = (\text{row } i \text{ of } \mathbf{A}) \quad \leftrightarrow \quad (\mathbf{A}^\mathsf{T})_{|i} = {}_{i|}\mathbf{A}$$

$$\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 1 & 3 \\ 4 & 1 \end{bmatrix}; \quad \mathbf{A}^\mathsf{T} =$$

$${}_{i|}\mathbf{A}_{|j} = {}_{j|}(\mathbf{A}^\mathsf{T})_{|i}; \qquad (\mathbf{A}^\mathsf{T})^\mathsf{T} = \mathbf{A}; \qquad {}_{j|}(\mathbf{A}^\mathsf{T}) = \mathbf{A}_{|j}$$

Symmetric matrices $\quad \mathbf{A}^\mathsf{T} = \mathbf{A}$

$$\begin{bmatrix} 3 & 1 & 7 \\ & 2 & 9 \\ & & 1 \end{bmatrix}$$

5 Vectors, row matrices, column matrices

$$(1, \quad 3, \quad -10,) = \begin{pmatrix} 1 \\ 3 \\ -10 \end{pmatrix}; \qquad \text{but} \qquad \begin{bmatrix} 1 & 3 & -10 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 3 \\ -10 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}; \quad {}_{2|}\mathbf{A} = (2, \quad 3,) = \begin{pmatrix} 2 \\ 3 \end{pmatrix}; \quad \mathbf{A}_{|1} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} = (1, \quad 2, \quad 4,)$$

When enclosing a list of vectors in square brackets, we create a matrix whose columns are those vectors.

$$\big[{}_{3|}\mathbf{A}; \quad {}_{1|}\mathbf{A};\big] = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}; \qquad \mathbf{A}^\mathsf{T} = \big[{}_{1|}\mathbf{A}; \quad {}_{2|}\mathbf{A}; \quad {}_{3|}\mathbf{A};\big]$$

6 Linear combinations of rows and columns

Linear combination of columns

$$\begin{bmatrix} \diamondsuit & \clubsuit \\ \heartsuit & \spadesuit \\ \diamondsuit & \clubsuit \end{bmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = 3 \begin{pmatrix} \diamondsuit \\ \heartsuit \\ \diamondsuit \end{pmatrix} + 4 \begin{pmatrix} \clubsuit \\ \spadesuit \\ \clubsuit \end{pmatrix}$$

MATRIX $\times$ ***vector*** $=$***vector***

Linear combination of rows

$$(1, \quad 2, \quad 7,) \begin{bmatrix} \diamondsuit & \clubsuit \\ \heartsuit & \spadesuit \\ \diamondsuit & \clubsuit \end{bmatrix} = 1\,(\diamondsuit, \quad \clubsuit,) + 2\,(\heartsuit, \quad \spadesuit,) + 7\,(\diamondsuit, \quad \clubsuit,)$$

vector $\times$ **MATRIX** $=$***vector***

Linear combinations $\longrightarrow$ $\boxed{\boldsymbol{a}\mathbf{B} = (\mathbf{B}^\intercal)\boldsymbol{a}}$

7 Vector times matrix

Remember that ${}_{i|}(\mathbf{A}\boldsymbol{b}) = ({}_{i|}\mathbf{A}) \cdot \boldsymbol{b}$; hence

$$(\boldsymbol{a}\mathbf{B})_{|j} = {}_{j|}(\boldsymbol{a}\mathbf{B}) = {}_{j|}\Big((\mathbf{B}^\intercal)\boldsymbol{a}\Big) = \Big({}_{j|}(\mathbf{B}^\intercal)\Big) \cdot \boldsymbol{a} = (\mathbf{B}_{|j}) \cdot \boldsymbol{a} = \boldsymbol{a} \cdot (\mathbf{B}_{|j})$$

Rewriting rules

$\boxed{{}_{i|}(\mathbf{A}\boldsymbol{b}) = ({}_{i|}\mathbf{A}) \cdot \boldsymbol{b}}$ and $\boxed{(\boldsymbol{a}\mathbf{B})_{|j} = \boldsymbol{a} \cdot (\mathbf{B}_{|j})}$

Thus, if we omit the period, we can simply write:

${}_{i|}\mathbf{A}\boldsymbol{b}$ and $\boldsymbol{a}\mathbf{B}_{|j}$

8 Matrix multiplication: **rows times columns**

Consider $\underset{m\times p}{\mathbf{A}}$ and $\underset{p\times n}{\mathbf{B}}$, then:

$$\boxed{{}_{i|}(\mathbf{AB})_{|j} = ({}_{i|}\mathbf{A}) \cdot (\mathbf{B}_{|j})}$$

Proof.

Remember that ${}_{i|}(\mathbf{A}\boldsymbol{b}) = ({}_{i|}\mathbf{A}) \cdot \boldsymbol{b}$, hence

$${}_{i|}(\mathbf{AB})_{|j} = {}_{i|}\Big((\mathbf{AB})_{|j}\Big) = {}_{i|}\Big(\mathbf{A}(\mathbf{B}_{|j})\Big) = ({}_{i|}\mathbf{A}) \cdot (\mathbf{B}_{|j})$$

$\square$

Thus, if we omit the period, we can simply write:

${}_{i|}\mathbf{AB}_{|j}$

9 Matrix multiplication (**by rows**)

Consider $\underset{m\times p}{\mathbf{A}}$ and $\underset{p\times n}{\mathbf{B}}$, then:

$$\boxed{{}_{i|}(\mathbf{AB}) = ({}_{i|}\mathbf{A})\mathbf{B}}$$

Proof.

Let's see that the j-th components are equal:

$$\Big({}_{i|}(\mathbf{AB})\Big)_{|j} = {}_{i|}\Big((\mathbf{AB})_{|j}\Big) = ({}_{i|}\mathbf{A}) \cdot (\mathbf{B}_{|j}) = \Big(({}_{i|}\mathbf{A})\mathbf{B}\Big)_{|j}$$

so ${}_{i|}(\mathbf{AB}) = ({}_{i|}\mathbf{A})\mathbf{B}$. $\square$

Thus, we can simply write:

${}_{i|}\mathbf{AB}$

10 Matrix multiplication (**by rows**)

Each row of **AB** is a linear combination of the p rows of **B**

$$\begin{bmatrix} 2 & 1 \\ 3 & 8 \\ 4 & 9 \end{bmatrix}\begin{bmatrix} 1 & 6 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 12 \\ 11 & 18 \\ 13 & 24 \end{bmatrix} \quad \text{where} \quad \begin{cases} \begin{pmatrix} 2, & 1, \end{pmatrix}\begin{bmatrix} 1 & 6 \\ 1 & 0 \end{bmatrix} = \begin{pmatrix} 3, & 12, \end{pmatrix} \\ \begin{pmatrix} 3, & 8, \end{pmatrix}\begin{bmatrix} 1 & 6 \\ 1 & 0 \end{bmatrix} = \begin{pmatrix} 11, & 18, \end{pmatrix} \\ \begin{pmatrix} 4, & 9, \end{pmatrix}\begin{bmatrix} 1 & 6 \\ 1 & 0 \end{bmatrix} = \begin{pmatrix} 13, & 24, \end{pmatrix} \end{cases}$$

11 Transposing a product of matrices

Since

- $(\mathbf{A}^\mathsf{T})_{|j} = {}_{j|}\mathbf{A}$
- $\boldsymbol{a}\mathbf{B} = (\mathbf{B}^\mathsf{T})\boldsymbol{a}$

it follows that:

$$\boxed{(\mathbf{AB})^\mathsf{T} = (\mathbf{B}^\mathsf{T})(\mathbf{A}^\mathsf{T})}$$

Proof.

$$(\mathbf{AB})^\mathsf{T}{}_{|j} = {}_{j|}\mathbf{AB} = (\mathbf{B}^\mathsf{T})\big({}_{j|}\mathbf{A}\big) = (\mathbf{B}^\mathsf{T})(\mathbf{A}^\mathsf{T})_{|j}.$$

□

The product of a matrix with its transpose is always symmetric

Library nacal for Python

Review the implementation of matrix algebra operations in the nacal library for Python that accompanies the course: Section 1.3 of the documentationSection 1.3 of the documentation (or study the code directly).

`https://github.com/mbujosab/nacallib`

You will see that the code is a literal translation of the definitions presented here; however, there is not a single line of code that describes the properties we have demonstrated in these three lessons. It is not necessary! Definitions imply properties (as we have theoretically verified with the proofs from these lessons). Verify with examples that all the properties hold. You should work with the Jupyter notebooks corresponding to the first three lessons.

Questions of the Lecture 3

Please complete the exercises from the book.

(L-3) Question 1. Calculate the following matrix products (in the indicated order) **EF**, **FE** and $\mathbf{E}^2$

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & 0 & 1 \end{bmatrix}; \qquad \mathbf{F} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & c & 1 \end{bmatrix}.$$

(Strang, 1988, exercise 34 from section 1.4.)

(L-3) Question 2. Indicate which of the following statements are true and which are false (in the latter case, provide a counterexample).

(a) If the first and third columns of **B** are equal, then the first and third columns of **AB** are also equal.
(b) If the first and third rows of **B** are equal, then the first and third rows of **AB** are also equal.
(c) If the first and third rows of **A** are equal, then the first and third rows of **AB** are also equal.
(d) $(\mathbf{AB})^2 = \mathbf{A}^2\mathbf{B}^2$.

(Strang, 1988, exercise 10 from section 1.4.)

(L-3) Question 3. Let the vectors $\boldsymbol{a} = \begin{pmatrix} 1, & -2, & 7, \end{pmatrix}$ y $\boldsymbol{b} = \begin{pmatrix} 3, & 5, & 1, \end{pmatrix}$.
Compute the following products

(a) $\boldsymbol{a} \cdot \boldsymbol{a}$ (b) $\boldsymbol{a} \cdot \boldsymbol{b}$ (c) $[\boldsymbol{a}]\,[\boldsymbol{b}]^{\mathsf{T}}$

(Strang, 1988, exercise 3 from section 1.4.)

(L-3) Question 4. Write the 2 by 2 matrices **A** and **B** whose elements are $a_{ij} = i + j$ and $b_{ij} = (-1)^{i+j}$. Multiply them to find **AB** and **BA**.
(Strang, 1988, exercise 6 from section 1.4.)

(L-3) Question 5. The product of two lower triangular matrices is again a lower triangular matrix (all its entries above the main diagonal are zero). Confirm this with a 3 by 3 example, and then explain why this fact follows from the laws of matrix multiplication.
(Strang, 1988, exercise 12 from section 1.4.)

(L-3) Question 6. Let **A**, **B**, **C**, **D**, **E** and **F** .

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & -1 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & -2 \\ 0 & -1 & 4 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & -1 \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} 0 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$$

Compute (*in particular, note that* $\mathbf{EF} \neq \mathbf{FE}$*!*)

(a) $\mathbf{B} + \mathbf{D}$ (b) $2\mathbf{E} - \mathbf{F}$ (c) **AC**
(d) **BC** (e) **CB** (f) **ACD**
(g) **EF** (h) **FE** (i) **CEF**

Strang, G. (1988). *Linear algebra and its applications*. Thomson Learning, Inc., third ed. ISBN 0-15-551005-3.

Strang, G. (2003). *Introduction to Linear Algebra*. Wellesley-Cambridge Press, Wellesley, Massachusetts. USA, third ed. ISBN 0-9614088-9-8.